

ActInf Livestream #020.2 ~ The Emperor's New Markov Blankets

Source: [YouTube](#) **Playlist:** Active Inference Livestreams (Paper discussions)

Duration: 1:55:07 | **Uploaded:** Unknown **Transcript method:** auto_caption

hello and welcome everyone to actin flab this is actin flab live stream number 20.2 this is our second group discussion on the paper the emperor's new marco blankets and today is april 27 2021 welcome to actimflab everyone we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at our links on this slide this is recorded in an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome and we'll be following video etiquette for group live streams like muting if we're not going to be talking and raising our hands so that we can hear from everyone we're down here on the bottom of the slide in the second discussion for the emperor's new markov blankets and closing out april with this really great discussion so check out this link rb.gy/kvnpyc if you want to see what's upcoming for the different streams the goal of today is really just to continue this awesome exploration and learning and discussion that we've been having and really appreciative that the authors are here to be talking about it with us so we'll be able to ask some cool questions and anyone who's watching feel free to ask a question in the live chat and we'll relay it to the group today we're just going to be going wherever we need to go we'll start with introductions and then if people want to call out a specific figure or a specific question uh or especially reference a quotation in the text or just something that they've prepared that would be really great so we'll just go around and introduce everyone and say hello feel free to just introduce yourself however you'd like and give a short check in and then after we have an introduction round we can return to these warm-up questions and anyone can feel free to chime in on one of these questions such as what they're excited about so i'm daniel i'm in california and i will pass it to shannon hey everyone i'm shannon i'm usually in california but i'm in south dakota for the pandemic and i will pass it to sweden hello i'm stephen i'm in toronto and i'm doing a practice-based phd through the solomon school of applied psychology and i will pass it to scott hello um scott david from seattle washington and at the university of washington applied physics lab and uh

we're working on synthetic intelligence and i will pass it to wait i lost my list who hasn't
chatted yet uh dean have you had a chance hi uh i'm dean i'm from calgary i'm retired and i
just like hanging out here because i learned a few things and i'll pass it to chris hi i'm chris i'm
a postdoc at the rural universitat in bochum in germany but i'm joining from berlin today and
i'm one of the authors on the paper and i'll pass it to joe uh thanks chris i'm jo dewhurst i'm a
postdoc at the munich center for mathematical philosophy also one of the authors on the paper
and my background's in philosophy of neuroscience and computer science primarily i'd say
thanks i'll pass together hi everyone uh happy to be here uh again i'm yada raiber also one of
the authors of the paper i am currently based in amsterdam although i'm supposed to be in uh
sydney australia um yeah uh happy to be here uh today today's a birthday at the dutch king so
it's a special day happy birthday okay i think that's everyone with the introduction so anyone
please feel free to raise your hand and start off some of the warm-up questions are what is
something that you're excited about today could be related to the paper or tangentially what's
something that you liked or remembered about the paper or from our discussion last week and
what is something that you're wondering about or you'd like to have resolved by the end of
today's discussion maybe another question would be what kind of momentum or what kind of
next steps would make sense you know emerging from this whatever we end up resolving or
not in the next conversation where will that take us in our research or in our practice journeys i
guess so joe and then anyone who'd like to speak uh thanks um do i need to lower my hand
again or is that done uh it's just good good great uh cool so i missed the discussion last week
so some of this may have already been covered but one thing which relates a little bit to
something we were discussing just before we went live here and i think is kind of um a
broadening of the points we wanted to make in the paper which i which i wonder about is uh
what what the if if active inference can be kind of arbitrarily applied to any given case given
the right kind of setup and parameters and so on um what does this mean for its future
application is this so far as it's too general or is this a strength insofar as it can be used
anywhere really um and this this i think is kind of lurking underneath some of the points we
make in the paper that's a more general cool so can active inference apply to anything and
everything would that be a strength or would that be a weakness awesome question any other
thoughts anyone wants to raise your hand yet steven i'd be interested to um learn more about
the way that the pearl blankets and the fristen blankets um interpretations might be useful in
applications and how that might inform you know the way that modeling's done so that that's
kind of exciting for me so that's one thing that i'm i'm looking forward to great question as

well the um distinguishing different kinds of blankets and even pulling back to like what kind of metaphors will we use and how will we relate technical terms to different settings different practices those are some big questions we want to address scott and then anyone else yeah i'm interested in all in both of those questions or all that aspects and then also the frizzle of the paradox of those two questions so it's a chicken and egg question of first and enough first of active inference in a sense right we're observing something we're projecting it out as a teleology onto the world and then asking if it's a good teleology right so it's so in a way it's like if you go for david bowm idea that all reality is first a thought that includes the realities that we don't create entirely but that we project so to one of the questions i have so as a i was a practicing lawyer for 30 years or 25 years ish and when i see active inference i think contracts and so i think of it as biomimicry of what we can do socially and project further with real intention to control systems so that so what i'm wondering is and i think both are valid but because so ultimately is it a notion i'll use the word biomimicry but obviously i don't mean in the physical but from us to learn from deep time what we can do as biological organisms to better organize ourselves despite our social elements so to get back to our biology in a way and and maybe organize ourselves a bit better and more resiliently with reference to that anyway it's a that's the frizzle i'm interested in is the both the cause being upward and downward with regard to active inference thank you cool and connecting the micro and the macro it's sort of sometimes it's laid out abstractly like multiple spatial scales that are nested within each other or time scales but then when we talk about action and we have the example of sort of the person in a maze and you take a step back so that you can go around that's the kind of settings where we might want to apply active inference where the initial action might be non-linear or away from the goal in a sense but it's part of a policy trajectory that actually gets you where you want to be so and the other the other part of it is with the markov blankets being a bayesian notion because of the beige then you say oh when does bayes when is bayes effective well that becomes a pretty broad application right because i always say to people to get them in the space i always say if you're in a dark forest with no flashlight you don't run full speed ahead you stick out your hand in front of your face right and that's basically active as i say you don't you know you want to know what's in front of you before you run into a tree right so anyway good stuff thank you any other yep yellow and then anyone else yeah so uh i think last week we ended uh discussion on shannon's point on what it means to have a folk book psychological form of active inference and i think it related to the uh to the tension between needing to have a very deep mathematical understanding of an increasingly kind of

developing field of mass and applying these kind of more broad set of ideas uh to a particular kind of case so i thought this was a a point to explore further and and think about how this would look like this uh this more folk psychological way of understanding active inference using the concepts maybe without the nitty-gritty details of mathematics shannon yeah um i was hoping that we could bring that up today and maybe ask joe's take um and also sort of if we're if we're talking about something like a fristen blanket and that's supposed to be a real demarcation of a system or a pearl blanket which is supposed to just be how we describe the system um then maybe in this you know folks psychological notion of active inference or talking about how a phenomenon appears to this person in a crowd versus another person in a crowd where both phenomena are real but depending on which one you decide to model you'll get a different picture of how the crowd's dynamics are changing and we could model that mathematically but just as easily someone in the crowd could have an informed notion of what it is that they're receiving and how it seems that the markov blankets however they are are emerging around the crowds in certain ways and you could start having a discussion that maybe you can you know translate from the scientists doing the fancy math and the person in the crowd who's just existing thanks very interesting steven than anyone else yeah just building on that i think i'm also curious because there seems to be there's two sides there seems to be this primordial soup which is referred to in the paper as well and the variational free energy and how that can self-organize um and then and that happens at smaller distances and then these longer distances where you've got expected free energy because the organism is seen into the environment with sensory information and often it's used in these partially observable markov decision processes for modeling that and i'll be just curious if those two domains of modeling which seem to come up and i know there's more but those two seem to come up a lot in the literature how that pans out maybe with some of these questions of you know is there an intuitive feeling of how the morphology is moving towards a target which could be that deeper sort of knowing of the body and then there's this kind of partially observable decision process which is kind of cognitive sort of looking in the world and which ones are more or less like markovian in some ways and how does that affect how we try and uh play with them which is or model them or play with them which i think shannon was saying in a way is in a good way is exploring these ideas phenomenologically thank you scott and then anyone else the stevens notion got me thinking about the notion of glass in glass they frequently talk about first order second order third order order um and so you have that amorphous material it's semi-amorphous solid right so it's not really crystalline not really

liquid so it has these orders that are discernible through different framings of maths and it made me think about stephen when you were saying that cognitive glass in a sense that we have this orders of order that we project onto disorder and so we those teleologies ultimately roll up into things like sovereignty which is seen as perfect order in the abstract right the sovereign does not need to ask for permission or forgiveness because it is the quintessence the fifth element right it is not earthly so we have this metaphysics that happens at the bottom or top of the turtle stack depending on which direction you're going and so and i think that's kind of nice here hey blue i think that's kind of nice here when we look at order and abstraction and the last point about it is that you know kandinsky the artist said that violent societies yield abstract art and so one of the things i always wondered is is abstraction the reverse of violence so if we look at sovereignty as being a total abstraction as i just described maybe sovereignty is also total violence and mills said sovereign has the monopoly of legitimate violence so anyway those are few ideas about maybe we're looking at cognitive glass here with different orders of order that we can project onto the system and maybe that helps us when those when we look at that order as levels of abstraction as you get further from the first and second order into the third and fourth order maybe it starts to be something that starts to yield a more projective or teleological element thank you thank you scott dean and then anyone yeah so i i wasn't part of the the tile formation last week i was kind of in the background and then lost it but i did go back and watch the the whole thing and i'm hoping that uh chris has a few more things to say today about construal because um that to me is kind of the the inflection point and he was starting to talk about some of those things and the separation between uh what did he have model and target i think at the end of the day um if we were to give a lay person or some i think the term folk has been used i think if we give them a mark off blanket and a pearl blanket um a lot of people will go how does that change my basic question which is how do i get the world i think that's what all of us here are trying to do and i think for a lot of people who aren't even aware of what a markup like it is or free energy principle is they still want that basic question answered so i'm really interested to find out today how the paper moves us to a place where that becomes easier that the obstacles to getting the answer to how do i get the world suddenly flatten and become i don't know more familiar thanks dean and a thought on the cognitive glass and what you just said there so framing it in big terms about sense making sense making about the world as well as about active inference is really helpful and scott when you mentioned the cognitive i thought about oobleck it might be called different things different places but it's cornstarch and water and so when you slap it really hard it's like a

resilient boundary and your hand comes away dry and it's like you're hitting a piece of concrete but then if you go very slowly it kind of is like a liquid so it's an amorphous solid like the glass now the boundary between the air and the oobleck is something that is real about the world and so there's a distinction there but it's almost like depending on how the boundary is crossed not even like two different types of things they're both touch but it's actually the dynamics of the stimuli that dictate whether the response is going to be very stubborn and more like a solid or whether it will be very giving and again it's not like it's light versus sound to different sensory inputs this is different quantitative differences within one interface modality and it just shows how even when there's a clear boundary in the world we think like between um two phases still the behavior of the interface can be very dependent on the type of interaction so scott and then anyone else made me think is it schenerian all the way down or do you have myriad schenerian vectors intersecting and we see that as complex stephen then anyone else i think you could also see this question around the glass because glass at a certain way of freezing or becoming solid it doesn't crystallize it doesn't break up it you can still see through it so you've got this question of it being part of the medium through which information can pass because you can see through it but there's also that question around that goldilocks zone when if you want something to rearrange and be the material like happens in the primordial soup kristen talks about there's a you know too lower temperature it sort of just becomes a solid too higher temperature it's going to dissipate and start to so you've you've got a bit of the overlap between you know internal external solid liquid um map territory it's kind of interesting yes different phases of the mind for sense making too rigid or too fluid so um joe then scott and then anyone else who'd like to speak uh thanks yeah so this this just made me think that quite quite a few of these points for me at least come back to a kind of central thing so so shannon's question about folk psychology which i'm very interested in or kind of folk inference um questions about teleology and uh questions about you know whether whether it's just something simple all the way down which makes it look like it's complex um to me makes me think there's an important question here about the role of um perspective and kind of viewing of these systems going on and and the way i see a lot of a lot of the more teleological things that we project onto these systems um some even even like a boundary perhaps it isn't really something we should uh take to be in the system that is rather just something we're uh kind of modeling the system as having so so it comes back to my target model question as well um so just just a central question here of what we take to be really in the world and what we take to just be part of our understanding of the system of a world cool thank you scott and then

anyone i love the conversation because everyone is moving so nicely and easily and with such uh facility back and forth between the metaphysics and the physics and appropriately and i don't i don't mean this is not a diminishment right because that we're we're demarking those edges one of the things here that occurred to me in the last couple of notes is the notion of seed crystals because we are talking about phase changes in system behavior i think there's very often a threshold kind of trigger type of thing that happens and so that notion that stephen was bringing up about phase seems interesting and that was implicit in what you were saying about cornstarch as well and so one of the things i think is interesting is what do those triggers look like and how are those encoded and in so when you're talking about popular encoding versus of some biological encoding so let's say my i'm triggered into anger but then that anger takes on the symbol of a certain person in a political party or whatever right there's different stages of phase change that are happening there of solidity that are being offered to that glass-like structure of order that are being set up by those externalities and it seems like one of the things we just did a paper on misinformation with a group and one of the things that's so interesting to me is it's not a single event misinformation it's not a binary where oh i got misinformation how did i how could i have prevented it like you know getting hit on the head with a stick there's so many levels at which you can address misinformation and intercede and intervene and this feels like one of those exercises that perhaps we can start to identify interventions for information integrity by looking at where might we put in seed crystals to put in phase changes at stages of that process to redirect it in ways that are more socially or productive or whatever the goal is of the group thank you thank you scott steven and then anyone else yeah i think this discussion also it points to an important consideration it brings in the territory and the environment because a lot of these phase changes in the world out there but also maybe unlike say the enamel of our teeth which is maybe like glass in a sort of what you would call an equilibrium state okay it's at a low energy state it's just there right i don't expect it to keep moving that much but we're talking about these non-equilibrium steady states so you've got this kind of micro states which things can revisit from which some are more probable from which there's attractor states which becomes these macro states which then the organism at different scales is visiting so this is interesting because there's both going on you know i think sometimes everything's non-equilibrium steady state but you know i'm holding the computer here it's pretty thermodynamic equilibrium object you know i'm i'm in that world even if my body has a lot of non-equilibrium dynamics going on so that maybe plays out in some of these questions thanks steven yep a lot of uh parallels analogies cross mapping between information

and thermodynamics and it's unsurprising i guess because there's a lot of groundwork there but it really is a fruitful intersection maybe we can just remind ourselves of some of the questions that we had from last week and then we can just continue walking along so anyone just raise a hand so uh these were like loosely structured by topic but we had big questions how much mathematics is required to understand or work with free energy principle and active inference and just how shall we speak and work clearly on this issue then we had a few questions on a philosophical side we recalled the papers of mel andrews and anes hippolyto and thomas vanesse which were a footnote in this paper as sort of relevant contemporary work so definitely on the philosophical side i'm sure we could um hear any thoughts about what next steps are on that line of research and this question of how to think about blankets for groups you know if it's groups all the way down then what's wrong with throwing blankets over groups but what kind of blankets and what kind of groups also blue if you want to introduce and give any thoughts that's cool too and of course questions about what's real or not um we had a few themes arising about again about math and about how there is um there's multi-directional dialogue with active inference and other fields so what happens when we cross over between different areas scott and then anyone else so i want to talk about markov flags for a second and what i mean is the flag of amsterdam with the three x's my understanding is from a tour guide that the x's stand for fire plague and flood and so if that let's assume that for a minute and the idea was if you see that flag you better follow the club rules on preventing fire plague and flood because that's what we do here in this area in amsterdam okay so to me that is a markov blanket because what you're doing is you're saying hey we are projecting this out come join the club we are doing beijing analysis on how to prevent cholera and floods and fire in the city limits and that's a the signal is there on top of every building that is has an owner who conforms to the rules and so i went i wondered if we can do a markov blanket analysis on the amsterdam flag as a markov flag expressing and saying all ye who come here right it's like diogenes he's looking for an honest man i'm looking for plague preventing and flood preventing people here so to me that's a projection out in in in any other kind of active inference we're projecting out through a flag instead of sticking our hand out to search for a tree so that's active and then i'm doing regulatory enforcement which seems to me to be a body that is responding to non-conformity among its members so could you comment on whether that's a possibility in terms of projection out of social elements of markov blankets possible any amsterdam resident is welcome to respond uh shannon and then anyone else who raised their hand um but yeah so now we're talking about something like human of cumulative

culture and our evolution of signs and symbols and how co-construct meaning in space so maybe the flag if you don't know what it means it's just the flag um you might not even know it's a flag if you see it painted somewhere it's just something that looks cool and has some shapes but if you have been raised or taken a tour i guess um in the culture surrounded by the symbol surrounded by the meaning all of those small instances of negotiating meaning and like we talked about active semantic inference i can't exactly remember what that conversation was last week but this seems like it could be similar sort of adding meaning and if the flag doesn't mean to the person you're interacting with what you thought it would mean to them you take action in your world to make it mean the thing to them that it means to you um it's an interesting thread i like that idea also one thought on the uh flag before we go to an amsterdam resident is like a blanket you think about under the blanket or inside the blanket like there's a person or a blanket goes on a bed but a flag is also fabric and it's flying in the wind and the borders it delineates are not on one side of the air versus the other side of the air but as the flag is animated by the niche through winds otherwise it's just on the flagpole it delineates a social barrier or social um demarcation which is then cultured in a intersubjectively negotiated experience just like you brought up shannon so yellow and then stephen all right thanks um it's fascinating uh i'm not sure i i i know this story yet from the three crosses i i think i have a different origin for it but maybe this is the tour guide so i'm not quite sure um i i think this race is also points to to one of the questions uh joe was writing in the beginning right which is in turn like suppose we can give this analysis of the flag of amsterdam at the markle blanket and we can give an analysis of uh the origin of life in terms of market and and lots of things in between right so what what what does this multi-applicability of uh micro blankets say it could one it could point to a very deep self similarity that runs through nature at all possible skills uh or or could mean that the toolbox is is in a sense to cause grains to to so that it applies to everything but but but you can't really make uh the relevant differences right i mean in a sense you would like to think through what are the differences indeed through a kind of uh uh signs of group memberships in in in social situations versus uh more biological science versus a kind of self-organization self-organization in physics and i think that that's that's kind of the tension between the kind of applicability to everything uh and and how particularly you would like your your theory to be thanks so steven dan scott than anyone i mean this is also brings that question i think like we just mentioned there where we put things in a markov blanket where things are part of this uh external states um hidden states internal states because in some ways from the markov blanket perspective the sensory states of the epithelia that the

kind of sensory organs is what's been stimulated and it's inferring an external state in i.e the flag and that's having an additional inference that the flag has got this meaning another level so um i'm not even sure what the whole thing's called in some ways because is the blanket because the blanket is dependent on connecting with external and internal but it's normally just those that part of the of the whole active inference dynamic so i think that that becomes interesting is where what type of uh sensory input information is within the blanket biologically and maybe that isn't quite the same as what is thought of when we start talking about it and that might be good or maybe a problem thanks so scott then chris than anyone this is uh this is fantastic this discussion so um i realized all of a sudden i thought oh wait a second i just realized that some of these folks may actually be empiricists and may believe there is a testable reality out there i didn't because i'm not that person so i i made a living rhetoric and a persuasive speech right as a lawyer right so for me it's all projection so i realized i was i was too diminishing of the question of whether there exists something out there to test and i was looking at and this whole thing to me this has been a candy store since a few months ago where i learned about active inference because to me it's all um about avoiding avoidable harms among humans and to the extent that active inference is universally uh assertable then it has a rhetorical power that's huge right so if we can get people thinking they're surrounded by active inference blankets and then i say you know you don't have an active inference blanket yet in the way you're doing business with other people in your supply chain or you don't have active inference blanket yet in the way that you're interacting among nation states or you don't have active inference yet in your food insecurity supply chain or your cyber security all these other things that i'm working on right now contact tracing it's information systems where because they're not within the same model then they have conflict and those most of those are avoidable harms because they're just humans working out stuff that we were separated for a hundred thousand years when we migrated out of africa and now we're getting back together and trying to compare notes on what we came up with so the question or this assertion is that it's interesting i to me it's not that troubling a question of whether these exist or not and i'm and i've read lots of philosophy and i've been down the road there but ultimately it's the the i say yes and on on that question of whether it exists or not to me it's a provisional truth that's functional sufficiently functional so i'm not no longer curious about whether it has an existence separate from the human mind so that so take my comments with a salt lick not a grain of salt but a giant block of salt because i've realized i'm an old old person and i just don't care if it actually exists separately and i'm not saying that to say this conversation doesn't

matter it's awesome but it's that that the dynamics of rhetoric are such that we can employ the thing to do good whether or not it's real so scott has litigated realism versus instrumentalism he's serving as the judge jury executioner lawyer um chris and then joe then blue yeah thanks a lot so um i think there's a like there's a there's a useful way to to think of the flag metaphor so to speak and to bring it a little bit closer maybe to to also what scott was saying now and and the paper itself so the question that the paper is trying to answer right it's kind of to what to what extent is the pristine blanket construct or the markov blanket construct as used by fristen a conventional thing so you know there are different like you you can have different views of course about the metaphysics of mathematics and so on and so forth but generally there's some agreed rules about how maths works or certain branches of mathematics work and that's where the mark of blanket construct comes from right it comes from statistics and there's a clearly defined way of using it so the convention for using the mark of blankets is very clear it was introduced very clearly by judea pearl in his article in later his book at the break of 80s and 90s and there's no kind of mystery or you know there's no question about how it should be applied productively so whether or not you you think that science is after capturing the reality as it is or whether you think science only gives us approximate truths or some kind of you know approximate uh models of one of of what's out there that that doesn't matter what's really important for us i think and maybe the guys can also speak to that is that for mark of blankets there's a clear convention for how to deploy them and it's very clear what they are doing and what they can't give you whereas for the way fristen applies the notion first of all he kind of softens the convention the convention is less codified we can think we can say it's less clear to what extent they apply where and similarly it's less clear what you can get out of it right so this is the um this is kind of the problem so so stephen for example mentioned something about you know boundaries and the fact that whenever there's an interaction these ma mark of bl sorry these blankets let's leave the mark of part of the name out of it these blankets appear and or they can be found there and it's not entirely clear whether according to pristine there will be found at every uh in every place where we have an interaction between two systems for example or something like that right because the the the the rules uh in which they are applied are not as clear as as in the case of the original markov blanket construct and i just wanted to also to follow that up on this very quickly in the version of the paper that's under review now we actually realized that perhaps in the draft you've read we were a little bit uh too harsh on tristan because he does impose certain rules about for example how to individuate different kinds of nodes but there's still a question of kind of what what it means to individuate

these nodes in a certain way thank you thank you joe thank you joe if you wanted then blue and scott i'm fine for now actually thanks thank you blue and then scott so without specifying markov blanket or pearl blanket or fristen blanket i want to touch back on what yellow was saying about group memberships and also about uh what shannon was saying about cumulative culture so i think this is like a really interesting notion of um like a markov blanket or a blanket overlying a group and and the affordances and like what it means to be part of that group like age specific affordances or socioeconomic specific affordances under a given blanket and how and when this like cultural or group membership prior gets like instilled and also like do blankets then overlap like i'm of this race and of this age and of this you know so so these overlapping blankets i just think it's a very um idea about group membership and cumulative culture and we might know that we're part of this group but we don't know all of the specific underlying backstory about why we're treated this way or were afforded these certain things i just think that's a very interesting way to think about it thank you blue scott and then anyone else and also anyone who's watching lives welcome to ask us a question so it's so interesting couple of things blue on what you just said so i do a lot of sewing and with the notion of quilting is really interesting in what you just said because they're actually a bunch of quilting museums that do ethnographies of quilts and so it really comes back around you had this quilt which reflects a person's history like when they did the hiv quilt each square represented a person right so you actually have a physical membership instantiated in this blanket and then the blanket being the symbol of the community being knit together i mean all these knits probably not the thing you want to do in a quilt but anyway okay the the thing i was gonna mention in response to joe i think was joe's comment before about the oh no sorry it was um it was kristoff's about the um application and the alteration from the paper that notion they have this um this something that's running as a scientific a statistical concept that is brought out in the world one of the things who as i mentioned before i work in an applied physics lab now applied physics is really funny because you're applying physics so that seems like maybe it has something to say about what's going on here and i and i realized well what you do in applied physics and what we do in our program in specifically in our program we take technically feasible systems so let's say that we could describe something as technically feasible the chris the fristen notion is technically feasible and you can apply it in scientific context but what i do in my program and what the lab does generally is then we take the stuff and make it ready for adoption in the field so in specifically in my program i take technically feasible systems and i test them for whether they're bolt reasonable and bolts is business

operating legal technical and social so if something is technically feasible like the orion project where they were going to use nuclear bombs to loft weighty objects into space that's totally technically feasible but it would be unreasonable socially because you'd pollute the entire atmosphere with radioactive isotopes right so it works in the lab it's theoretically good terrific so what i'm my notion is maybe what's going on here is the applied physics applied statistics and when you get it out there in the field it's it can be seen we can call them compromises or we can call them realities or we can call them affordances or we can call them beijing adjustments of scientific empirical views right which when it gets out there in the reality maybe the folk understanding is the reality of physics in the mind of the of humans maybe if the mind exists in society and the brain's just an antenna tuned into it anyway just a couple thoughts to play with thanks scott so stephen and then anyone else who has a question yeah i'd be interested christoph just further in what you said about what does it mean to individuate a node in a particular way in this idea of nodes and i suppose at this larger scale um i'm seen with these partially observable markov decision processes being used in some of the modelling they don't necessarily from what i understand talk nodes in that way but at the primordial soup kind of cellular level they seem to often talk about modes in that maybe in in the similar way or not but i'd just be curious what you think thoughts are about what it means to have a node what kind of node that might be and what's similar or different between different ways that this is implemented yep go for it chris and i have figure 2 up or let me know if you want to go to a different figure uh sure thanks a lot i mean those two the last two comments and the questions even really match nicely together i think because so you know i mean you can we can talk about variables right i mean in a in a graphical model variables are represented as nodes basically so what but what i of course meant is the differentiation of the differentiation between active states and the passive of or perceptual states inner states uh yeah exactly well you can also use the loop but yeah internal states external states um these things right so it seems to me uh and i think i mean not only to me it seems to all of us that there's a certain degree of arbitrariness with regards to what is what i mean at least in the draft you've read it seemed like that to us we then learned that after really reading for a lot of tristan's papers very carefully we actually kind of extracted the implicit rules for for how to how to you know label these variables as active or passive in a graphical model but this is still a very arbitrary distinction one which is uh like the rules by which you individuate so i i used individuation here maybe i shouldn't have used the word individuate the way you identify the role played by a given variable that's what i should have said previously is that the rules that

that give you this identification are still very arbitrary and they are not actually anchored in anything very substantial and this goes back to what scott said about the feasibility and applicability because of course you can take this kind of way of partitioning a system you can build a robot with you know some ai system which will be built on the principles of the free energy principle and pristine's way of thinking about graphical models and it will be perfectly you know good at doing certain things that you build it for the problem is that when you're building a system for a particular purpose as an engineer you kind of hold all the cards you make all the decisions you know you decide like how to what like what kind of active states will the robot need or something like that and that's very that's a very different perspective so to speak from this perspective of you know let's call it the what then it calls the evolutionary r d uh department right so an evolution doesn't really plan things evolution just throws things that the wall and whatever sticks kind of stays and goes on to reproduce and all the other things kind of fall back and um in this sense it's it's very it becomes very problematic when we think of like pristine's grand project of explaining you know life as we know it for example as in the paper from which we took this diagram that's on the screen now because it's it slowly becomes a little bit suspicious whether we can actually whether the the the way he labels these nodes and the the kind of the rules for labeling the nodes will be the same on for example different levels of organization stuff like that right so so whether this will be applicable uh to many different biological systems in the way that he hopes it will be or not thanks and just to to clarify um you all wrote that spectral graph theory is used to identify the eight most densely coupled nodes which are defined as the internal states so for those who think that given this soup of interacting particles which are all the same kind they're all following the same simple rules yeah eight densely coupled ones okay those are internal and then there's some downstream consequences with which nodes influence which nodes but it's hard to argue that choosing the eight densely coupled nodes is simply you know true or internal when you can look at what it is and ask whether there's a different way to partition it so joe and then scott and then anyone yeah thanks thanks daniel that's a very clear way of putting that point um so just to pick up again on something chris chris mentioned briefly there is a further individuation problem i think which is about how to identify which which nodes you have in the first place or which which variables you have and this is a kind of growing question so so you can always you know look inside a node and split up in some other nodes or some other subparts and this this might not be a problem just that active inference suffers from it's going to be any any any kind of modeling uh system is going to have this kind of issue but it is going to make a

difference to where the boundaries are drawn and what what kinds of what kinds of things you get so that's just to say that there's there's already a kind of human uh involvement in the model and just choosing what scale you pitch your nodes at thanks joe uh scott and then dean yella and stephen so in that notion love it this is great and that clarification about the nodes being assigned in a sense contextually in a given system state um it seems like if we start to that may be a pathway to revealing the self other ch or starting to unpack some self other challenges so maybe a node that is once characterizable as internal then characterizable as external is revealing something about the paradox membership in groups that the in fact it's not the node that changes but the sensation of whether itself or other so you start to take that always paradoxical this is that hofstetter stuff and everyone else who talks about when does a group become a when does a group form when does it have an internal voice when does it speak an extra voice all that stuff and so maybe that analysis and that reassignment possibility starts to get into that and maybe that also informs blue's notion before about the group you know kind of the again that i'm calling it now the markov quilt as of that last conversation but that group identity blanket where it's patched together maybe those things are like irving goffman and things like that or mirror neurons where you want something to be recruited by others as an internal state you project it to them as an external state intrinsically and so the communication you're inviting others to encode their markov blankets similar to yours in a sense or something like that so that is exciting to me that slipperiness of assignment there feels like something that offers a pathway to analyzing those dynamics of group and individual identity thanks thank you scott so dean yellow and stephen and blue yeah um so back to construal and the question of how do i get the world i think a lot of the conversation today has centered on maybe a more clunkier expression which is how do us get the world either us as flag conformists or us as red dots versus cyan dogs um so even pristin stephen shared a video where he was talking to the santa fe institute even pristin said that things get a lot more interesting when we get to the combinatorial level and they're not the same as when we're talking about how do i get the world that there is there is a difference when complexity arises so can we hold up pearl and mark off blankets and get past the ore of this or that because we recognize that it's a little bit different when how do us get the world that's a construal question and so that's what i'm kind of curious thanks dean really powerful and a big difference between how do us get the world how do we get the world how do i get the world how do they get the world these are like really markedly different questions so yellow stephen blue um so i was thinking getting back to blue's question and shannon's questions on group membership but

maybe if somebody else wants to uh well i i i i just get talking on a group membership um so i mean to what extent one the kind of marker blanket formalism applies to to groups uh i think it's an interesting one and i think that's one thing that really matters here is is really how you conceive of a group right i mean you can lead like a group of people i mean we are a group of people that are spread out all over the world and you could think like a group in a more narrow sense like a flock of birds or a these are entities in the same kind of spatial temporal location that coordinate their their behavior with one another and you can think of as a predator arriving also as a perturbation to that system as a whole right so in the way that entity is coordinating it's it's it's it's dynamics with with the rest of the environment you might actually think that no this is a group that actually too weak the active inference principles do apply and you might in that sense might think as well that the group as a whole somehow has a as a as a as a marker blanket um i think it's that another analysis is more difficult to give if you think uh group memberships uh also kind of the graded group memberships right i mean uh i think blue was talking about uh yeah we belong to multiple groups at the same time uh and not actively acting on being part of that group but just as a kind of label that i would uh endorse uh when you ask me i think there the application of markup blankets is a lot more difficult and at most more metaphorical so i just want to say so exactly how to understand groups there is is is is is important if if shannon wants to understand kind of crowded damage or so that that might be more applicable and if you want to really understand kind of social membership and graded membership and membership of different groups at the same time then uh yeah i find it more difficult to understand how kind of the active influence framework can be applied to those kind of groups thanks so steven blue scott yeah i think so sticking with that group idea that there could be the the idea of morphology and the amount of constraints and affordances in the the whole environment body organism dynamic that's constraining and where maybe there's not that constraint so i'm thinking i know someone that she's trying to do some markov blanket meditation processes i'm kind of thinking if someone's meditating starting to just sense their states in a maybe a a way that slowly is like a cloud that self-organizes in a way i could see a more plausible way where sort of something like not obviously anything the same as this kind of markov blanket in a soup could be emergent however when you're in a you know a more real cognitive environment where you've got these priors and and the only way we're able to access it anyways through self-reported data which we a much higher level of dimensionality it may be is is only in some of the temporal processes that we can get some analogies like but i think that yeah it may be very different at different scales so the way that constraints play out

i'm wondering how we think about constraints um and not having constraints nice question steven are the constraints hooked into this phrasing and then or where are the constraints and how does that shape how we model things across different scales so blue and then scott and then anyone else so hopefully it won't go on too long but i want to touch back on what stephen just said and also what um yela said you know when i think about how group membership goes into active inference like how can we relate these two things i think about like specifically what is the likelihood that i'm gonna take a ride on a yacht right like so my socioeconomic status doesn't really afford me the opportunity to ride around in yachts very frequently so i'd be very surprised and i also live in the desert so i'd be very surprised if i was riding around on a yacht at three o'clock this afternoon like that would that just doesn't go into my model so perhaps our group membership um goes into you know construction of of the prior um and our model in ways that are maybe layered right like you know an onion maybe um and so then i want to touch back on what scott said because this is what really made me raise my hand about group alignment and we were starting to talk with casper hesp when he was on the stream about um inferring the states of others so i think this group alignment i wonder about this a lot and and uh specifically with respect to communication which we've talked about in relation to active inference and cooperation and you know building a collective intelligence so when we think about you know whether the earth is flat or not right like this at one time was like a hotly debated topic and maybe still is but um this is like there's been a convergence right so we all now align our like most of the people in the world agree that the earth is round so i wonder about this time to convergence and this has been used to measure information transmission and can we similarly move it through estimate collective intelligence so like when we there's a time to convergence a swarm or this might also represent like a phase transition right so there's a time to convergence and if it happens rapidly like the idea is readily adopted that now the earth is round so i wonder if these divergent views that we have as people in groups in divergent groups in a society that do they hinder our collective intelligence or do they like actually assist it because without having divergence we could never actually come to the one true answer so is it resistance that hinders the evolution of intelligence or is it um i just it's something to think thanks blue um scott if you're go ahead yep i just repeat the question you asked right there at the end yeah repeatedly yeah so i just wonder our um the divergent views right that we hold in society about whether the earth is flat or round do does this divergence help or hinder our collective intelligence and the evolution of intelligence overall or is it resisting adoption of a new view right like so perhaps there's there's

resistance to change even when we see something is correct like climate change we can clearly see the effects of climate change but there's a huge resistance to changing so is it this resistance that hinders our collective intelligence like what where is the evolution of our collective intelligence helped or hindered by divergence and views or resistance to adopting new views sorry goes along yeah shane you want to give a thought on that and then scott yeah um so if we were going to talk this would be like whether a system resisting it's a non-equilibrium steady state and like moving to a different steady state um i guess that's a question i don't i like i've been seeing pop up in a lot of the papers recently is how can you define not every system is going towards like a minimized free energy gradient or not every system is trying to maintain a stable state like systems dissipate and re-emerge and dissipate and re-emerge um so i don't know is that just a way to re-ask that question but talk about something a bit more physical than than collective intelligence um and i guess collective intelligence maybe is like collective agreement on a set of of ideas or norms or something rather than like intelligence like smarter than another is what we're maybe talking about here so how does that system arrive on this steady state of what we think that the world is thanks shannon and there's the convergence on shared generative model on shared mindset but then it made me think of ants colony relocation and over evolutionary time colonies are shaped to appropriately relocate but not instantly and not just choose the first um you know first aid coming back with exciting nest location you can't go for the first one so how to actually bear with the process of distributed computation or distributed intelligence knowing that it looks like disagreements at multiple levels especially when it's not just as simple as ants relocating but cultural values and beliefs and everything like that can i jump in again about aunt yeah yeah oh yes there was a new paper and i don't know who was by i saw it on twitter um but it was about ants they travel across a scaffold and it's either it's like flat and so they just walk or they tilt it and so some ants start to fall down and they start to build this scaffold um and then when they build a scaffold ants can all walk over it so they're supporting each other um and i'm thinking um i might be diverging from our markov blanket topic here but i'm thinking um that collective intelligence is like as long as the world looks flat as long as the world is easy for us you know to navigate then we can all go about our day we don't really need to try to converge with anybody but as soon as the world starts to be disrupted slightly so we turn that scaffold folks start to fall off like then we need to build something build a shared collective notion of reality and then then the convergence matters and so that could be something like as soon as there's a perturbation to the system then it's useful to try to recognize boundaries or try to have some sort of

intervention i think scott was mentioned to have a boundary that's suitable and helpful and productive and that could be where it's fine to diverge all you want as long as everything's simple and then as soon as it gets complicated or we start to like have harm for people then we need to have a system that can result in some sort of convergence happy for someone to bring me back on topic it's it's a great point and i almost interpreted as how do we help those at the fringe who might be being harmed how do we make that the stimuli and um for realignment and for re-scaffolding rather than keep on cutting off more and more oh the failing parts of the bridge they gotta go they gotta go well now we have no one on the bridge instead of seeing that those on the fringe and can be part of the recovery so scott kristoff and steven so shannon that was right in the middle of the discussion there's no you don't have to bring back anything that was perfect so admiral rogers the former head of the nsa once said don't let any emergency or disaster go unexploited and the idea is that the disaster is that situation people act in a disaster you know hurricane sandy whatever people you know starbucks they were putting out power strips for people to share whatever right so it's exactly that and the rhetoric the projected projection rhetoric can cause a state change right the anticipation of a disaster when people are actually experiencing they go oh my god this might happen to me or neighborhood watch right people kind of you know they go in and they want to protect property and things so i love that that point and bringing it with the other points before i think it was kristoff that was asking maybe somebody else about you know that active inference in a group and does it dissipate in the activity if you're just a member of a club and it's lying dormant is that really describable with active inference and it occurs to me you know if you look at group dynamics it may still be usefully described that way and here's here's how i would think of it if i uh this is irving goffman's stuff that i i project myself out as i want to be perceived by others so if i join a club i may join a club because i want to be like i used to be part of an entomology listserv i've never been a this is not about enterman's cakes but insects you know but i was part of a lister back in the old listserv days and i just was fascinating i sort of wanted to be involved in it right so i was projecting myself out and i started making these connections with folks so in a way you can de-risk and leverage by being a member of a group even passively right i can i'm extending out my feelers out there into the world by having that group membership i'm de-risking by having those that group available to me as me being a member if it's a group that you know so so i wonder if the dissipation of activity itself is an appropriate gauge of whether you're a member of a group or maybe in certain leverages and certainty riskings there's a functional engagement that's maybe more passive by the individual

but still may stand for something in the overall goals of that entity that's that's one point and the other one other point is the the leveraging the risk in groups i keep coming back to eukaryotes and eukaryotes you know we it's a better deal we got multi-cellular got a lot more options we got mitochondria we got chloroplasts we got all the good apparatus instead of having just these compartments where you kind of do stuff in a single cell organism right so it's a better deal and so maybe part of this thing with the groups this goes back to blue's notion is is it are we resisting i think it was a blue maybe but are we resistant to is this resistance or facilitating the group membership what's going on here and you know the divergences and the differences and i started thinking about thermodynamics first law if you don't have if you have isothermal you can't perform cannot perform work if everyone has equal information markets don't have trading so there may be a benefit to not having to having divergence right and maybe in fact um the that reality is there isn't a facilitation or resistance it just is and that paradox is what we're is the reality we're seeing is that it doesn't facilitate anything there's no goal orientation individuals may have goals but that's a more bounded goal than the super system of biology anyway just a couple of thoughts that i gleaned from that last bit and also just to remark once again what a fantastic conversation thanks scott so kristoff then stephen um yeah so i just wanted this well what did i want to say uh i guess two things really so one thing goes back to blue's comment like the fact that the fact that if you have some kind of a complex network there will be multiple markov blankets in there so it's completely feasible to understand you know membership to certain groups or something like that or some kind of relationship structures to be overlapping in different ways so i think it's conducive to this but i think that we should not like we're very focused now on this idea you know blankets as um as a way of intervi like of thinking about some kind of boundaries again between groups perhaps or between different kind of units of organization and we're not like it's it's it's a it's always useful i think to kind of go back you know zoom out a little and not think about the blankets because you have the blankets whenever you have some kind of a network of you know variables um interconnected in some way but what's really interesting is that in many cases the behavior you will get from these systems if you project them onto the social domain will be very surprising so one really surprising finding is something called the solman effect it's with a z and double l and it's a finding that shows that if you have bayesian agents interacting in a network if the network is fully connected they won't actually come to a consensus about some true belief so so imagine you have a network of fully connected network of bayesian scientists and they're testing two hypotheses right and they're testing these hypotheses they're increasing

their credences and they're communicating their credences with each test to their neighbors and what's really surprising is that the talmud effects shows us that these scientists will not actually converge to uh to a singular credence in in the true outcome what is also surprising is that a less connected network will actually eventually converge to the to the to kind of to all members having a very high credence in you know most likely outcome so this kind of i just wanted to kind of throw it out there because i think that it's you know we're thinking now intuitively about these blankets as membership to groups and stuff like that but the kind of interactions we might get from using this kind of formalism and projecting it to a social domain might be quite surprising right i'm not saying it will be useless because perhaps the talmud effect to go also back to what has been said about the flat earth situation and you know the belief in the roundness or flatness of the earth the solemn effect actually tells us that in a hyper-connected society like the one we are living now there it's even rational agents might fail to converge to the to the you know the most likely hypothesis right and that actually is something interesting that it's in a way that solomon effect predicts the appearance of certain conspiracy theories in a hyper-connected society uh given the demographics for example of flat earth uh belief it has been declining for decades until the advent of internet so it only came to a sharp rise with the with with social net with social media when people started to communicate in a much more interconnected way so i'm not saying you know thinking about these systems in a social domain is bad no i just wanted to bring this let's say anecdotal effect to to say that uh the the kind of behaviors we will see these systems uh converging towards might be very surprising right or we we might they might not work exactly the way we they do on like lower scales of organization thanks for those points so steven and then anyone else yeah that's some good points there actually and i think this is where active inference um and the mark the fristen blanket has some advantages over the pearl blanket if it's about what action it can there's a risk you see the information you start to have this thing that's all information driven rather than action policy driven because it and by keeping it with the action policy um for instance one of the things about having us on the planet in the solar system which we used to have well now we know in a galaxy in a milky way blah blah blah is that people have this sense of look we almost meant to look at ourselves from space now which is really only a proviso of modern technological perspective people might want to be able to stay grounded in some way there may be actually a psychological challenge with having to take this outer space perspective of us on a little ball going around the sun you know so there could also be an action policy perspective here around flat earth gives you a way to re-ground

yourself because it's like well i'm on a flat earth so i don't have to take that i don't have to go and imagine myself in space looking at myself so there's a lot of interesting aspects to the action policy side once you start to have this scaling and that that could be interesting um to to keep that in mind thank you steven joe and anyone else uh yeah thanks um just something you said for steven uh when you say there's an advantage to the frist and blankets over the pearl blankets do you mean an advantage you know for real systems making use of these things or an advantage for us as modelers this is something that i really worry about sometimes that we slip into thinking about the practical benefits for a system of of these constructs whereas what we're really concerned with is the benefit it's for accurate and and in that context both might be good tools but when you use one tool or the other will depend on what you're trying to do with them i would say more for how we look at it as a modeler although the actual thing gets called a system and then has this kind of perspectival process i think it's good to keep it in so that action doesn't get lost and it doesn't purely start to become information yep great now that that makes sense and um i wasn't necessarily accusing you of this error but sometimes i do worry that people uh people get mixed up over who who's getting the advantage i guess you know who's who's benefiting here but but if if you mean just that it's important to keep to keep action as part of the picture and part of our models that that's a fair point yeah and one contribution of this paper and we look forward to the um revised and published version as well is it's almost like if all we know is it's like hey hand me the screwdriver well there's different shapes there's different sizes if you have the hex it's not going to work for the phillips and different countries or different applications need different screwdrivers so now we can with these adjectives ahead of blanket we can start to differentiate hand me the phillips screwdriver okay you're talking about the fristen blanket you're talking about the pearl blanket and so we can be specific now because it's almost like enough air was pumped into markov blanket metaphor it was becoming too big of a tent too big of a blanket and so now we can start to be really specific and and address those kinds of questions when are we talking about utility for the system when are we talking about utility for the modeler which of course maps very closely to um the distinguishing in uh that you raised with inference with an inference within a model so scott oh yeah if any of joe if you want to give a thought on that and then yeah can i can i just i'm really so we're already pleased with the reception this paper's had this particularly my reactive inference community and just just the reason you you said that i one thing we were worried about but people have seen is that there is meant to be this more positive message that these aren't necessarily bad tools they're just

different tools for different kinds of cases so i'm i'm really glad that people have seen that and kind of taken that as helpful message it's it's the yes and approach to the literature it's not blood sports it's not a knock down drag out with citations we can just build upon each other and uh come to more useful and good ways of communicating so scott and anyone else yeah just two points one along those lines in in law one of the most effective strategies for disambiguating something is saying what it isn't right and so even if you don't know what it is you can at least start to say what it isn't and so that's a handy notion along the lines here and as last couple of comments we're observing we had this thing out there that was very exciting and we kind of everyone was playing with it and said oh yeah and it's the same things happened in privacy the word privacy is just carrying water for all sorts of anxieties right now the word trust has all sorts of precision so the problem with language is we got a lot of subtlety and a lot of interactions interactions are increasing exponentially through exponential increase in interactions globally is an assertion that i make at least and if you think about that we still have a vocabulary that's the same as it was way down the exponential curve of interaction increase so how do we capture our subtleties in our interactions right how do we expand our vocabulary what's exactly what we're talking about here is we take words we use them by analogy in other areas right we've already talked about blankets well these aren't really blankets at all and in fact we get hijacked by the notion of blanket we've already seen that it's starting to use those is it a blanket or is it a membrane well is it semi-permeable well a blanket's not really semi-permeable you know that kind of stuff right so it's fascinating that we have this is the the appropriate and um very exciting challenge to have to the taking it and run running with it idea early on in disambiguation and it means that there's a lot of energy which means there's a lot of need for whatever it is that's being served by the concept and so what you're as modelers this gets kind of interesting kind of gets into a manhattan project kind of ethics of science right because as modelers you do have responsibility and you're discharging it here by saying hey what does this mean for society you know should we develop nuclear energy unfettered with regard to whether it can be used as a weapon right because this is this is powerful stuff here and so that raising of the notion and the questioning of whether it's appropriate to apply these things socially is going to come up very quickly as people do things like the rhetoric that i'm talking about and start to apply it politically and start to recruit biases using these concepts that will be seen as a nefarious application of a scientific and mathematical statistical concept and so your paper and the questions you're raising now are starting to lay that foundation of pause and saying to people in the future hey don't be so fast

to run with this and just take a minute to think about the implications what isn't this right and so starting to ask that question can be as important and sometimes more important than what it is thanks thank you there's a question from the chat sean kelly wrote are there special cases of markov blankets that shield internal states a markov shield so would any of the authors like to just give a response to that yeah yellow and then go ahead stephen i i can give it a try um of course this really uh depends on what you mean by shielding uh right and uh i think in the paper and i mean in the paper we use a statistical notion of shielding that is pretty much equivalent to conditional independence right so if if you have three nodes and there's one intermediate node such that knowing the intermediate node will will will mean that knowing the third node doesn't not mean anything doesn't give you anything for for knowing your first node uh then that's a notion of shielding in this in in the sense that the second node kind of epistemically shields off uh the the activity of the at the third node from from whatever happens at the first node and this is an epistemic notion of epistemic and statistical notion of shielding and also that one right i mean shielding has a lot of kind of more everyday kind of connotations that are necessary not necessarily can be congruent with this physical uh notion but if you mean a statistical notion then yes the very definition of a markov blanket is that the things inside of the blanket are given the marker blink it's shielded off from the rest of the world in that statistical sense thank you that was what i was thinking insulation like a heat insulator is shielding it and so that is kind of what these nodes are doing statistically and then we're nuancing what kinds of of categories these nodes might fit into and that's a key difference between pearls phrasing and fristen's phrasing is the extent to which nodes are classified into different kinds so steven and then anyone else yeah i think that question about how things and when we scale up is really important and in predictive processing and the modeling you know you have this hierarchical model of things going up to higher levels and yeah there could be this implicit suggestion that each of those steps or somewhere up there it's going to other size blankets but it may not be that blankets are needed as much um to go another layer it might be that there's some sort of process structure that is doing the job at lower levels it's a more active influence process and if i could give one metaphor there was a metaphor i'd put on about if there's a flag stuck in some mud and it looks quite stable but it's blowing in the wind okay and then i have another flag which is stuck in some mud it's actually the top of an ant hill and underneath the ants are all swarming to hold it still so and it's actually a little bit more still but actually the hole is more open so in the first case it's energetically thermodynamically more an equilibrium is being held in a sp in one place the second one you

think it's in more of a thermodynamic equilibrium because it seems more firm but actually there's a whole swarm of ants underneath trying to hold it in place and but none of them have got that intention but looking from the outside the system is holding the flag but from the point of view from the ants they're trying to enact some sort of action policy and i suppose i wonder if there's something similar between you know the system as seen from the outside as a state and the kind of swarming that's going on underneath and that the two can get mixed up sometimes thanks steven um always a great uh challenge an eternal question what are the ants trying to do but what you just pointed out there is is really it's related to this distinction they draw with these four different cases the difference between a generative model which is like the modeler's perspective on a something versus the generative process those are two different generative processes that are giving rise to the movement of these two different flags let's say there's two different processes underlying it those are realist claims about what is giving rise and then there's all different kinds of perspectives that can be taken um but if we mix up two people's perspectives or if we mix up a generative model a generative process it's a category error and if we're coincidentally correct it would be miraculous because it'd be a the wrong kind of thing that's being considered so it is extremely important to um prevent that kind of misinterpretation by really asking are we talking about the process degenerative process or the generative model and sometimes even just that distinguishing as well as this four-fold uh classification raised here these go a long way towards helping model so scott and then anyone else since we have europeans on the line i'd like to talk about kant for a second and i want to display my ignorance of kant um and the categorical imperative so i i i complain about the categorical imperative because i if can one person know enough have enough situational awareness so that their good acts can translate into goodness in the world and again maybe that's total misinterpretation of con or maybe just got it all wrong but in the same way i wonder about markov blankets and this notion of knowledge of an awareness so that because because that one of the problems we see in the world is even a good actor if they don't have good in information can do bad things like in the sustainable development goals of the united nations they put in a gujarat power plant was funded by a un based financing agency and it cost a billion dollars or whatever and they were crowing about the success of sustaining sustainable development goals of energy and development were being pursued and furthered but what they didn't talk about and they then got sued in in court over is the fact that the local fishermen and local population was pollution and so it poisoned the local waters and the local population with the pollution so they didn't do a sufficient analysis even in this situation where

they're trying to improve the world and listing out all these goals of the world that's their declaration that's their amsterdam flag of what they're trying to do with the u.n right so but nonetheless they pursued one goal at the expense of other goals and one of the things i'm interested in your thoughts you can go into contact one and also correct me on my bad county and non-interpretation there but um can you comment a little bit on awareness and knowledge we talk about group and individual markov blankets but can we talk a little bit about that notion about sharing of knowledge and situational awareness just as a survival strategy the ants were surviving in the um shannon's tilting bridge example before um which was a survival thing in stephen's flag projection that was a little bit more the answer a little bit smarter about their externalities than that one anyway but can you comment a little bit about how do we get how how might these analyses and these models and their implementations in reality and in the lab help us to understand the nature of group information float thanks thanks for the question so if anyone has thought yep gala go for it and then great i i'll give it a try i'm not sure if i can tie it all the way back to the categorical imperative but that's uh let's give it a go um so i think the first thing that the way i understand it uh the kind of active inference framework is really a theory of learning for a particular environments so in a sense organisms start to learn to adapt to the specific niche niche that they are that they are engaged in and then somehow slowly build up a particular model of that particular kind of niche that you can of course and that that model you can use also for hypothetical kind of situations or counterfactuals or or new kind of situations but there's no guarantee that this kind of previous uh uh i mean if you if you if you enter a new niche or if the world changes it's it's not guaranteed that your previous knowledge will be uh uh will be specific to the new situation and we'll be we'll work in in a new particular kind of situation um so i guess there that indeed the the active inference framework gives you a situated situation specific knowledge and not on the kind of categorical the domain general knowledge and so this doesn't guarantee indeed that if you use your or use your model to make inferences about new situations that that you will end up on on the right result and so that gap between what will be your best according to whatever kind of imperative you set for yourself in reality and how it works out that's always a gap in in in the knowledge that you have uh what you need is related to the difference between the kind of the causal dynamic of the world the generative process and the general model of the world that you that you have and so i get that that that's indeed why perhaps i mean speculating that that different kind of ethical frameworks might might might fit better in such a kind of approach rather than [Music] the other logical one that was very oh yeah go ahead chris chris uh yeah i

just wanted to follow up on this a little bit because we the the notion you know the the distinction between the generative model and the generative process was brought up before and now yella alluded to it and i think it's really important to keep in mind that the free energy principle so so the generals maybe let's start differently the generative process right is supposedly the process or the the the causal process that produces certain sensations in an organism and then the generative model is supposed to be on this framework producing predictions right which which kind of explain away in the bayesian sense uh the differentiations that the model is the the the organism is experiencing and the idea that's implicit in a lot of the literature is that you know you have the generative process on this side you have the generative model that's trying to mirror the generative process there's the sensory sheath in the middle and that these two these two things will somehow you know fall down to each other and they will be just perfect uh there will be some kind of a perfect resemblance or mapping between them um and that's not entirely it's not entirely clear that this is the most successful evolutionary successful strategy so the generative model only needs to recapitulate the generative process to the extent that it's successful for in a given ecological niche or even social niche to ensure the survival and reproduction of the organism right um and so so that's one thing that's i think important to remember about this that that you know the free energy principle from the get-go kind of does not necessarily produce agents that are perfectly that will be always um optimally informed about what's going on in their surroundings they just kind of really they they learn just enough to stay alive and of course the inclusion of active influence gives them an advantage over traditional bayesian agents because they can also go and try things out and learn in this way but it still give take us all the way so to speak uh to to to kind of transcend our model and and and and get to know the the process that that that's out there in the world and i think this is also true in the social domain in many ways so so in the social domain i think it's very plausible that the same thing happens and we could expect that the same thing will happen for i know groups for example uh where where groups will form certain models about how society operates or how the world is which will be optimal to their given epistemic situation and they're given social situation but will not be objective uh models of how the how society or the world operates thanks and what you just said there at the end about the cultural ways of viewing the world it's um known that many cultures their own name for their own country or homeland might be like this is the middle spot or this is like the main place it makes sense and it it's um it's unsurprising that it's that way but just like scott said we're all comparing notes that's the phase of our test that we're in and two people are

mapping to a different territory with the same relative egocentric map and so this question about how different egocentric perspectives and at which level uh those egocentric perspectives exist at how those come into harmony is really an question so stephen and then anyone else and yeah this is awesome times it's been a great discussion we have 20 minutes or however much more people would like yeah i'd just like to i really thought that was a really interesting point there about when other approaches are maybe better than taking an ontological approach and i suppose it i think and you're talking about the epistemological approach i suppose that you get into this question of whether it's a more fuzzy i'd be interested in your thoughts on like fuzzy logic and a more fuzzy based epistemology um rather than maybe the type of logical categorization of knowledge which ends up becoming ontological at some point and that there is this idea of ontological pluralism but they're sort of their way around this you know or third wave is that the third wave inactive consciousness has that this idea that the mark of blankets because it breathes with different regimes of attention you it's sort of trying to get around that problem but there is it doesn't necessarily say how they do that so just be curious yeah a bit more and and the other thing if it's tying in with other areas is like for instance they're talking about affect now a lot and affect is this kind of integration which then ties in in qualitative research where you have axiology which is kind of the the normally this thing which is sort of down there in arts-based methods but you know the the one where it's about how people feel about things well now that starts to become part of a lot of what we're understanding about the world so it could actually have quite an implication for participatory interactive kind of research methods and making them much more central in a way so i wonder any of your thoughts on that about about that from the authors thanks steven if anyone has any thoughts on that or can raise their hand yeah yell it go for it um i mean so so i guess what what we do in our paper is a little bit before that that step right so uh on the question of uh uh ontology phases versus epistemology should you go for a kind of ontological pluralism or not i think what you tried to say is luke here's a is a class of models that are typically used in literature as abstractions from a target phenomenon right and so if you want to enter if you want to understand those uh uh in kind of on the logical terms that that's that's fine but you need to give a story about how how that abstraction relates to the target phenomenon uh uh that is that that is there so either uh if you go to into metaphase you go from a kind of more uh platenese kind of root and you say well actually this mathematical model is not an abstraction from the target system but it actually nature at its jones is somehow mathematically structured um or you you say look no these are these are indeed

abstractions produced by us and they they're useful in predicting and interpreting the world uh but they are not themselves the the uh so in a sense we we kind of offer the the kind of dilemma to anyone who who wants to make a commitment or a choice there between epistemology or ontology or anthropological pluralism um saying like look you you can go the ontological route and uh of course this is the way more challenging interesting kind of exciting uh route but then you either need to bite this this this bullets that that these models are not really abstractions but somehow underlying reality or need to give some some other sort of story about how that how it works and uh i mean i'm i think personally i'm curious to see what what what the response from from the community will be uh which of those uh options people will take but i mean i don't think either of us is necessarily committed to one of those those which are not but just saying okay look if you want to do it then these are these are the rules kind of so to say maybe disagree with the rules that's fine as well but then give an argument thanks very interesting stuff anyone else have any thoughts uh yeah stephen and then anyone else go for it yeah i think that i i agree a lot with what you're saying there i think there's there are some gaps in um and and rather than saying there's a gap sometimes it's just filled with the markov blanket if that makes sense um if in doubt fill it with a markov and i've been really curious i'm actually looking a lot in terms of how organisms structure like the idea of parry personal space and and spatial inactive ecological approaches and and maybe there's a way that that space is actually um and there's a field called mental space psychology which has been a very fringe field um actually a lot of it based in holland and that um is uh it seems more plausible now with this so it was a much more practice based approach but um i thought i'd share that anyway because i'm presenting something this friday so um but it's i i do think there's something missing in terms of um mechanisms that extend out beyond the body into the space and it may not have to be um at all levels pristine's blankets in in the traditional sense shannon and then anyone else um i'm gonna talk about brains so i know we've been extending you know into the world into the body into the body and everything um and we've been talking a lot about dynamics and maybe how groups behave or how we feel like part of a different group um and earlier we super briefly touched on mere neurons um so these are neurons that fire when i observe an action the same as when i do an action and there's been a lot of work lately not necessarily on near neurons but on inter brain synergies or you might call them interbrain coherence or cross brain coherence where lots of dynamics in my brain are firing very similar to lots of dynamics in your brain um and this could be because they can work from mary hassan on neuro cinematics this could be because we're watching a film and

it's evoking similar responses because we're all attending to the same stimulus but the film that we're watching it could have a structure where the filmmaker sort of designs what you pay attention to um and so most people's gazes will fall in that direction um you'll have even more similar sort of inter-brain coherence or unstructured where you're just sort of filming a scene and there's maybe some people over here some people over there and your gazes will could differ a little bit more these structures could differ a little more um and this is all to say how like the film is the environment in this case if we're talking about the rest of the world being the environment or like us in this room being the environment our gaze is structured in a certain way by there's a blue box on my jitsi screen so it's sort of directing everyone's attention at my face um but there's also some lack of structure i don't know that we're seeing the same tiles in every space and there's different ways that if we interacted like if i interacted with one of you a lot in my daily life like maybe steve and i hang out and get coffee every day and we talk about something every single day so we have a propensity to pay attention to similar things in similar ways and we've been specifically talking about yella's idea and how much we agree or disagree with it or something so we are continually like drawing our gaze to yellow to see if he understands um or agrees with what we're saying like there might be this similar inter-brain coherence between stephen and i that's different between me and blue blue and i never talk we're not even friends um these are all hypothetical by the way um and these so a question it's not you know um in your paper necessarily but like if we're looking for markup blankets in the world one thing that we could look at is something like a inter brain coherence and that might be something real in the world or it might be a pearl blanket something that we use to describe how people interpret something the same or how their brain activities are synergistic in a certain way and if you can be empirical i know we were being rhetorical a lot of times but if you can be empirical and look at these signatures of either you know conversation dynamics or brain dynamics and can relate them to how the world is structuring them then that's a way that you can look for you can still keep looking for markup blankets in the world or find it useful um it's definitely i think what we keep coming back to when we're um these blankets as a metaphor these blankets is something real is that we can totally have both as long as we're specific about which one we're referring to thank you shannon yeller and then anyone thanks shannon that's really interesting and i think that there there there's there's a couple of things there right i mean and and in either one sense if two people have the same same background same experiences all right they might they might parse a particular kind of movie or scene in in in in a similar kind of way such that if you put them in in room separate

from each other you will indeed see a similar kind of neurodynamic going on in both both cases maybe tight indeed to attention where you pay attention your gaze etcetera um the other one is indeed kind of on-site kind of interpersonal synchronization that might be happening right and i know there's this kind of it's kind of second person neuroscience kind of approaches that that try to capture those kind of things in the experimental setting and see how how how in what particular kind of settings brainwaves synchronize and uh et cetera um and that kind of second perspective reminds me a lot of uh uh of the kind of discussion that's there in active inference on generalized synchronization right so these are the kind of examples that you get when as there's uh uh two coupled clocks standing on the table and is tend to synchronize after after a while so this is a very simple causal mechanism uh causal structure where synchronization just occurs over or over over time um and when i first started thinking about uh uh marco blankets i i used that kind of example as a kind of reduction that absurd him saying like look if if these things even have a marker blanket right then it's an empty uh uh empty notion now having worked with these with these philosophers of science here i think i need to be a bit more more more specific and indeed say look okay whether that thing has or has not a michael blanket that that that's that that's a different question but there's there is a point that there is a very uh uh simple structure of kind of causal dependence and causal independence such as if you would fix the table perfectly the things would not synchronize and if you would let it slide a little bit and wobble a little bit then they would they do synchronize so there is a kind of thing like this intermediate factor that mediates a causal relationship between those uh uh those two things now the question is that do you need to have a markup blanket for that does mark do market blankets add anything there or is it kind of a story about kind of causal dependence and causal independence enough i think that well causal dependence there and cults are independent such that if i would do this then there would be no synchronization i think that's enough and then what do you call that a kind of markov is is a second kind of question you need to translate it back to the kind of world of causal models and and see how they relate to them uh but i think there's yeah there's a clear case to be made that this is parallel the kind of cases you are describing and the kind of synchronization settings that are there in in this kind of couple clock examples thanks and one interesting aspect to bring it back to this discussion about the internet you know what kind of beliefs propagate culturally on what kind of connectivity uh just like shannon was saying with the video chat and with the live stream what somebody sees it's not just like they're sitting in a different chair at the theater and they're watching the same play or they're in separate movies

uh you know rooms watching the same movie and so cultural cues might lead somebody to look into a correlated position as another person but um when there's the customization it's like each person is in their own theater which takes it to another level because it goes even it's actually different inputs and the inputs can be targeted in interesting or new ways so it's kind of stretching the limits of our social cognition because it's not just a bunch of people asking what they just saw with a shared event it's like trying to come to a shared understanding when people are seeing different events so yeah there's some new fun things we're learning on the internet i guess well in the last couple of minutes if anyone has some thoughts i'd be curious to the authors or non-authors where do go from 20.2 where's our momentum or direction or curiosity one thing i'll bring up while someone's raising their hand is uh without uh buzzing in the middle of a conversation how can we be really clear when we're intentionally or unintentionally slipping the difference between a markov for pearl first and and generative model generative process with a model within a model we don't want to police subtleties of language but at the same time somebody can really build a bridge that leads someone to nowhere if they're unintentionally uh not sure the difference between these concepts so i'm just curious how do we promote uh rigorous and accessible norms around talking about these ideas recognizing that we're all like totally learning this and and formalizing the project around us yela and then anyone else uh yeah just a small uh uh note and i'm not i don't remember exactly who who mentioned this but uh at some point we talked about kind of two phases of research right and one is kind of taking a lot an id and running along pushing it along as as far as you can and then there's a new kind of phase where like okay now we ended up in a completely new new uh how how how are we gonna kind of substantiate this how can i make kind of formalizes a little bit and i think this is the the kind of the i guess over the last few years active inference knows the philosophy of active influence has kind of been just and also i mean also in my own earlier papers kind of been running as far as possible see how far we could get this and see what would happen as a kind of creative kind of kind of startup stage of of research and i think both in this paper and then a number of other papers that are currently kind of come coming out um uh i think this kind of more kind of consolidatory fake consolidatory phase is is happening where we're like okay but okay this is where we ended up does it really make sense need do we need to make some more fine-grained demarcations to to do we need to tweak the conceptual apparatus to to to have it all make sense uh and that it's a different kind of work but but i think that that's that's that's really important in the kind of for the next kind of phase for for for active inference uh research so i hope these kind of discussions can

contribute to uh uh to doing that kind of work we're consolidating bitcoin to the key 20 000 level you know let it rest before the next uh maybe whatever metaphor works for people but um any other thoughts because that's very nice yellow thanks everyone for sharing this was an awesome sequence and it really helped take our our lab and our discussion to another level we would always welcome any of you to come back to continue the conversation come for a guest stream come as a visitor to um be on any conversation you'd like so thanks again to everyone and for listening peace